

Package ‘GTAPViz’

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Title R Package for Automating 'GTAP' Data Processing and Visualization

Version 1.1.1

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Description An R package designed to streamline the processing and visualization of 'GTAP' model results. While primarily built for 'GTAP' simulations, it also supports other .har and .sl4 datasets. As an extension of HARplus, GTAPViz simplifies 'GTAP' data extraction and visualization, reducing manual effort and enhancing efficiency. With intuitive commands and automated adjustments, users can create, customize, and export high-quality visualizations for economic analysis and research reporting with minimal coding.

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Encoding UTF-8

RoxygenNote 7.3.2

BugReports <https://github.com/bodysbobb/GTAPViz/issues>

URL <https://bodysbobb.github.io/GTAPViz/>

Imports HARplus,

ggplot2,
dplyr,
tidyr,
openxlsx,
colorspace,
grDevices,
scales,
utils,
methods,
stringdist,
stats,
glue,
openxlsx2

VignetteBuilder knitr

Suggests knitr,
rmarkdown,
devtools

Depends R (>= 3.5)

Contents

add_mapping_info	2
auto_gtap_data	3
comparison_plot	6
convert_units	9
create_export_config	10
create_plot_style	11
create_title_format	16
detail_plot	17
get_all_config	19
get_color_palette	20
pivot_table_with_filter	21
rename_value	23
report_table	24
sort_plot_data	26
stack_plot	27

Index	31
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add_mapping_info	<i>Add Mapping Information to GTAP Data</i>
------------------	---

Description

Adds descriptions and unit information to GTAP data based on a specified mapping mode. Supports external mappings or default GTAPv7 mappings, allowing users to enrich datasets with standardized metadata.

Usage

```
add_mapping_info(  
  data_list,  
  external_map = NULL,  
  mapping = "GTAPv7",  
  description_info = TRUE,  
  unit_info = TRUE  
)
```

Arguments

data_list	A data structure containing GTAP variables.
external_map	Optional data frame with mapping information (must include "Variable", "Description", and "Unit" columns).
mapping	Character. Controls how descriptions and units are added: • "GTAPv7" (default): Uses standard GTAPv7 definitions. • "Yes": Uses only the information provided in external_map. • "No": Skips adding mapping information. • "Mix": Prioritizes values from external_map, but fills in missing entries using GTAPv7 definitions.

description_info Logical. If TRUE, adds description information to the data.

unit_info Logical. If TRUE, adds unit information to the data.

Details

The mapping argument supports:

Value

A data structure with added mapping information, preserving the original structure.

Author(s)

Pattawee Puangchit

See Also

[convert_units](#), [rename_value](#)

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

# Add mapping using GTAPv7 defaults
gtap_data <- add_mapping_info(sl4.plot.data, mapping = "GTAPv7")

# Use an external mapping file
my_mapping <- data.frame(Variable = c("qgdp", "EV"),
                        Description = c("Real GDP", "Welfare"),
                        Unit = c("percent", "millionUSD"))

gtap_data <- add_mapping_info(sl4.plot.data, external_map = my_mapping,
                             mapping = "Mix")
```

auto_gtap_data

Process GTAP Data Automation with Flexible Output Options

Description

Processes GTAP data from sl4 and har files with options for exporting and preparing plot-ready data.

Usage

```
auto_gtap_data(
  experiment,
  input_path = NULL,
  output_path = NULL,
  sl4_suffix = "",
```

```

har_suffix = "",
process_sl4_vars = NULL,
process_har_vars = NULL,
mapping_info = "GTAPv7",
sl4_mapping_info = NULL,
har_mapping_info = NULL,
sl4_extract_method = "get_data_by_dims",
har_extract_method = "get_data_by_var",
sl4_priority = NULL,
har_priority = NULL,
sl4_convert_unit = NULL,
har_convert_unit = NULL,
decimals = 4,
rename_columns = TRUE,
region_select = NULL,
sector_select = NULL,
subtotal_level = FALSE,
plot_data = TRUE,
output_formats = NULL,
sl4_output_name = "sl4.plot.data",
har_output_name = "har.plot.data",
macro_output_name = "GTAPMacro",
add_scenario_ranking = FALSE,
rank_column = "ScenarioRank"
)

```

Arguments

experiment	Character vector. Case names to process.
input_path	Character. Path to the input folder; overrides project_path/in if specified.
output_path	Character. Path to the output folder; overrides project_path/out if specified.
sl4_suffix	Character. Custom suffix for SL4 files (e.g., "", "-custom").
har_suffix	Character. Custom suffix for HAR files (e.g., "-WEL").
process_sl4_vars	Character, NULL, or FALSE. Variables to extract from SL4 data: • Character vector: Specific variable names. • NULL: Extract all. • FALSE: Skip SL4 processing.
process_har_vars	Character, NULL, or FALSE. Variables to extract from HAR data: • Character vector: Specific variable names. • NULL: Extract all. • FALSE: Skip HAR processing.
mapping_info	Character. Metadata mode for variable descriptions and units. Options: "GTAPv7" (default), "Yes", "No", "Mix". See add_mapping_info() for full details.
sl4_mapping_info	Data frame or NULL. Mapping for SL4 variables. Must include columns "Variable", "Description", and "Unit".
har_mapping_info	Data frame or NULL. Mapping for HAR variables, structured as above.

sl4_extract_method	Character. SL4 extraction method: "get_data_by_dims", "get_data_by_var", or "group_data_by_dims".
har_extract_method	Character. HAR extraction method. Same options as above.
sl4_priority	Optional list. Required only when sl4_extract_method is "group_data_by_dims". Specifies priority rules for SL4 data grouping.
har_priority	Optional list. Required only when har_extract_method is "group_data_by_dims". Specifies priority rules for HAR data grouping.
sl4_convert_unit	Character or NULL. Optional SL4 unit conversion. Valid options: "mil2bil", "bil2mil", "pct2frac", "frac2pct". See convert_units for details.
har_convert_unit	Character or NULL. Optional HAR unit conversion. Same options as above.
decimals	Integer or NULL. Number of decimal places to round numeric values. Set to NULL to disable rounding.
rename_columns	Logical. If TRUE (default), renames GTAP codes (e.g., "REG" → "Region", "COMM" → "Commodity").
region_select	Optional character vector. Filters data to selected regions. Applies only to the "REG" column, which is fixed and cannot be modified.
sector_select	Optional character vector. Filters data to selected sectors. Applies only to the "ACTS" and "COMM" columns, which are fixed and cannot be modified.
subtotal_level	Logical. If TRUE, includes subtotal rows. Default is FALSE.
plot_data	Logical. If TRUE, generates plot-ready data and assigns to specified variable names.
output_formats	Character vector or list. Output formats for export. Valid values: "csv", "stata", "rds", "txt".
sl4_output_name	Character. Variable name to assign SL4 output. Default: "sl4.plot.data".
har_output_name	Character. Variable name to assign HAR output. Default: "har.plot.data".
macro_output_name	Character. Variable name to assign macro output. Default: "GTAPMacro".
add_scenario_ranking	Logical or "merged". Adds a numeric index for each scenario: • TRUE: Adds a ranking column. • "merged": Also prefixes experiment names with the rank.
rank_column	Character. Name of the ranking column. Default is "ScenarioRank".

Details

- To prepare data for plotting and generating tables within the GTAPViz package, the "Unit" column must be included in the output.
- When using the extraction method "group_data_by_dims", the corresponding priority list must be defined via the sl4_priority or har_priority argument. See [group_data_by_dims](#) for more details.

Value

A processed GTAP-formatted dataset with standardized structure and metadata, ready for analysis or visualization.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [convert_units](#), [rename_value](#),

Examples

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")

# GTAP Macro Variables from 2 .sl4 Files named (EXP1, EXP2)
# Note: No need to add .sl4 to the experiment name
gtap_data <- auto_gtap_data(experiment = c("EXP1", "EXP2"),
                           har_suffix = "-WEL",
                           input_path = input_path, subtotal_level = FALSE,
                           process_sl4_vars = NULL, process_har_vars = NULL,
                           mapping_info = "GTAPv7", plot_data = TRUE)
```

comparison_plot

Create Comparative Bar Charts from HAR and SL4 Data

Description

Generates comparative bar charts using GTAP data. Supports panel facets, split-by grouping, and fully customizable styling and export options.

Input Data**Usage**

```
comparison_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  split_by = "Variable",
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  invert_axis = FALSE,
  separate_figure = FALSE,
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  output_path = NULL,
  export_picture = TRUE,
```

```

    export_as_pdf = FALSE,
    export_config = NULL,
    plot_style_config = NULL
  )

```

Arguments

data	A data frame or list of data frames containing GTAP results.
filter_var	NULL, a vector, a data frame, or a named list specifying filtering conditions. For example: <code>list(Variable = c("EV", "qgdp"), REG = c("USA", "THA"))</code> .
x_axis_from	Character. Column name used for the x-axis.
split_by	Character or vector. • Column(s) used to split plots by (e.g., "REG" or <code>c("COMM", "REG")</code>). • If NULL, a single aggregated plot is generated.
panel_var	Character. Column for panel facets. Default is "Experiment".
variable_col	Character. Column name for variable codes. Default is "Variable".
unit_col	Character. Column name for units. Default is "Unit".
desc_col	Character. Column name for variable descriptions. Default is "Description".

Plot Behavior

invert_axis	Logical. If TRUE, flips the plot orientation (horizontal bars). Default is FALSE.
separate_figure	Logical. If TRUE, generates a separate plot for each value in panel_var. Default is FALSE.

Variable Display

var_name_by_description	Logical. If TRUE, uses descriptions instead of variable codes in titles. Default is FALSE.
add_var_info	Logical. If TRUE, appends variable codes in parentheses after the description. Default is FALSE.

Export Settings

output_path	Character. Directory to save plots. If NULL, plots are returned but not saved.
export_picture	Logical. If TRUE, exports plots as PNG images. Default is TRUE.
export_as_pdf	Logical or "merged". • FALSE (default): disables PDF export. • TRUE: exports each plot as a separate PDF file. • "merged": combines all plots into a single PDF file.
export_config	List. Export options including dimensions, DPI, and background. See create_export_config or get_all_config .

Styling

plot_style_config	List. Custom plot appearance settings. See create_plot_style or get_all_config .
-------------------	--

Details

Please refer to the full plot

Value

A ggplot object or a named list of ggplot objects depending on the `separate_figure` setting. If `export_picture` or `export_as_pdf` is enabled, the plots are also saved to `output_path`.

Author(s)

Pattawee Puangchit

See Also

[get_all_config](#), [detail_plot](#), [stack_plot](#), [create_title_format](#)

Examples

```
# Load data
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))
reg_data <- sl4.plot.data[["REG"]]

# Generate plot
plotA <- comparison_plot(
  data      = reg_data,
  filter_var = list(Region = "Oceania", Variable = "qgdp"),
  x_axis_from = "Region",
  split_by    = "Variable",
  panel_var   = "Experiment",
  variable_col = "Variable",
  unit_col    = "Unit",
  desc_col    = "Description",

  invert_axis      = FALSE,
  separate_figure = FALSE,

  var_name_by_description = FALSE,
  add_var_info            = FALSE,

  output_path      = NULL,
  export_picture   = FALSE,
  export_as_pdf    = FALSE,
  export_config    = create_export_config(width = 20, height = 12),

  plot_style_config = create_plot_style(
    color_tone      = "purdue",
    add_unit_to_title = TRUE,
    title_format = create_title_format(
      type = "prefix",
      text = "Impact on"
    ),
    panel_rows = 2
  )
)
```

convert_units	<i>Convert Units in GTAP Data</i>
---------------	-----------------------------------

Description

Converts values in a dataset to different units based on predefined transformations or custom scaling. Supports manual and automatic conversions for economic and trade-related metrics.

Usage

```
convert_units(
  data,
  change_unit_from = NULL,
  change_unit_to = NULL,
  adjustment = NULL,
  value_col = "Value",
  unit_col = "Unit",
  variable_select = NULL,
  variable_col = "Variable",
  scale_auto = NULL
)
```

Arguments

data	A data structure (list, data frame, or nested combination).
change_unit_from	Character vector. Units to be converted (case-insensitive).
change_unit_to	Character vector. Target units corresponding to change_unit_from.
adjustment	Character or numeric vector. Specifies conversion operations (e.g., "/1000" to convert million to billion).
value_col	Character. Column name containing values to adjust (default: "Value").
unit_col	Character. Column name containing unit information (default: "Unit").
variable_select	Optional character vector. If provided, only these variables are converted.
variable_col	Character. Column name containing variable identifiers (default: "Variable").
scale_auto	Optional character vector of predefined conversion rules: "mil2bil": Converts million USD to billion USD (divides by 1000). "bil2mil": Converts billion USD to million USD (multiplies by 1000). "pct2frac": Converts percent to fraction (divides by 100). "frac2pct": Converts fraction to percent (multiplies by 100).

Details

If both change_unit_from and scale_auto are provided, the function prompts the user to choose between manual and automatic conversion.

Value

A data structure with values converted to the specified units.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [rename_value](#), [sort_plot_data](#)

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

# Convert million USD to billion USD
gtap_data <- convert_units(sl4.plot.data,
  change_unit_from = "million USD",
  change_unit_to = "billion USD",
  adjustment = "/1000"
)

# Automatic conversion from percent to fraction
gtap_data <- convert_units(sl4_data1, scale_auto = "pct2frac")
```

create_export_config *Create an Export Configuration*

Description

Creates a configuration list for controlling plot export settings. This function provides auto-completion for export options.

Usage

```
create_export_config(
  file_name = "gtap_plots",
  width = NULL,
  height = NULL,
  dpi = 300,
  bg = "white",
  limitsize = FALSE
)
```

Arguments

file_name	Character. Base name for exported files. Default: "gtap_plots".
width	Numeric. Width of output in inches. Default: NULL (auto-calculated).
height	Numeric. Height of output in inches. Default: NULL (auto-calculated).
dpi	Numeric. Resolution for PNG export. Default: 300.
bg	Character. Background color. Default: "white".
limitsize	Logical. Whether to limit size. Default: FALSE.

Value

A list with export configuration parameters.

Author(s)

Pattawee Puangchit

Examples

```
# Default export configuration
default_export <- create_export_config()

# Custom export configuration
custom_export <- create_export_config(
  file_name = "regional_impacts",
  width = 12,
  height = 8,
  dpi = 600
)
```

create_plot_style	<i>Create a Plot Style Configuration</i>
-------------------	--

Description

Creates a configuration list for plot styling that can be used with GTAPViz plotting functions. This function provides auto-completion for style options while maintaining compatibility with direct list specification.

Usage

```
create_plot_style(
  show_title = TRUE,
  title_face = "bold",
  title_size = 20,
  title_hjust = 0.5,
  add_unit_to_title = TRUE,
  title_margin = c(10, 0, 10, 0),
  title_format = list(type = "standard", text = "", sep = ""),
  show_x_axis_title = TRUE,
  x_axis_title_face = "bold",
  x_axis_title_size = 16,
  x_axis_title_margin = c(25, 25, 0, 0),
  show_x_axis_labels = TRUE,
  x_axis_text_face = "plain",
  x_axis_text_size = 14,
  x_axis_text_angle = 0,
  x_axis_text_hjust = 0,
  x_axis_description = "",
  show_y_axis_title = TRUE,
  y_axis_title_face = "bold",
  y_axis_title_size = 16,
```

```

y_axis_title_margin = c(25, 25, 0, 0),
show_y_axis_labels = TRUE,
y_axis_text_face = "plain",
y_axis_text_size = 14,
y_axis_text_angle = 0,
y_axis_text_hjust = 0,
y_axis_description = "",
show_axis_titles_on_all_facets = TRUE,
show_value_labels = TRUE,
value_label_face = "plain",
value_label_size = 5,
value_label_position = "above",
value_label_decimal_places = 2,
show_legend = FALSE,
show_legend_title = FALSE,
legend_position = "bottom",
legend_title_face = "bold",
legend_text_face = "plain",
legend_text_size = 14,
strip_face = "bold",
strip_text_size = 16,
strip_background = "lightgrey",
strip_text_margin = c(10, 0, 10, 0),
panel_spacing = 2,
panel_rows = NULL,
panel_cols = NULL,
theme = NULL,
color_tone = NULL,
color_palette_type = "qualitative",
positive_color = "#2E8B57",
negative_color = "#CD5C5C",
background_color = "white",
grid_color = "grey90",
show_grid_major_x = FALSE,
show_grid_major_y = FALSE,
show_grid_minor_x = FALSE,
show_grid_minor_y = FALSE,
show_zero_line = TRUE,
zero_line_type = "dashed",
zero_line_color = "black",
zero_line_size = 0.5,
zero_line_position = 0,
bar_width = 0.9,
bar_spacing = 0.9,
scale_limit = NULL,
scale_increment = NULL,
expansion_y_mult = c(0.05, 0.1),
expansion_x_mult = c(0.05, 0.05),
all_font_size = 1,
sort_data_by_value = FALSE,
plot.margin = c(10, 25, 10, 10)
)

```

Arguments

show_title	Logical. Show or hide the plot title. Default: TRUE
title_face	Character. Font face ("bold", "plain", "italic"). Default: "bold"
title_size	Numeric. Font size of title. Default: 20
title_hjust	Numeric. Horizontal alignment (0 = left, 1 = right). Default: 0.5
add_unit_to_title	Logical. Append unit to title if applicable. Default: TRUE
title_margin	Numeric vector c(top, right, bottom, left). Default: c(10, 0, 10, 0)
title_format	List or function output. Title formatting options. Can be created with create_title_format(). Default: list(type = "standard", text = "", sep = "")
show_x_axis_title	Logical. Show or hide x-axis title. Default: TRUE
x_axis_title_face	Character. Font face for x-axis title. Default: "bold"
x_axis_title_size	Numeric. Font size of x-axis title. Default: 16
x_axis_title_margin	Numeric vector c(top, right, bottom, left). Default: c(25, 25, 0, 0)
show_x_axis_labels	Logical. Show or hide x-axis labels. Default: TRUE
x_axis_text_face	Character. Font face for x-axis labels. Default: "plain"
x_axis_text_size	Numeric. Font size of x-axis labels. Default: 14
x_axis_text_angle	Numeric. Angle of x-axis labels. Default: 0
x_axis_text_hjust	Numeric. Horizontal justification of x-axis labels. Default: 0
x_axis_description	Character. Optional description for the x-axis. Default: ""
show_y_axis_title	Logical. Show or hide y-axis title. Default: TRUE
y_axis_title_face	Character. Font face for y-axis title. Default: "bold"
y_axis_title_size	Numeric. Font size of y-axis title. Default: 16
y_axis_title_margin	Numeric vector c(top, right, bottom, left). Default: c(25, 25, 0, 0)
show_y_axis_labels	Logical. Show or hide y-axis labels. Default: TRUE
y_axis_text_face	Character. Font face for y-axis labels. Default: "plain"
y_axis_text_size	Numeric. Font size of y-axis labels. Default: 14
y_axis_text_angle	Numeric. Angle of y-axis labels. Default: 0

`y_axis_text_hjust`
 Numeric. Horizontal justification of y-axis labels. Default: 0

`y_axis_description`
 Character. Optional description for the y-axis. Default: ""

`show_axis_titles_on_all_facets`
 Logical. Show axis titles on all facets. Default: TRUE

`show_value_labels`
 Logical. Show or hide value labels. Default: TRUE

`value_label_face`
 Character. Font face for value labels. Default: "plain"

`value_label_size`
 Numeric. Font size of value labels. Default: 5

`value_label_position`
 Character. Position of value labels ("above", "outside", "top"). Default: "above"

`value_label_decimal_places`
 Numeric. Number of decimal places in value labels. Default: 2

`show_legend`
 Logical. Show or hide legend. Default: FALSE

`show_legend_title`
 Logical. Show or hide legend title. Default: FALSE

`legend_position`
 Character. Legend position ("none", "bottom", "right"). Default: "bottom"

`legend_title_face`
 Character. Font face for legend title. Default: "bold"

`legend_text_face`
 Character. Font face for legend text. Default: "plain"

`legend_text_size`
 Numeric. Font size of legend text. Default: 14

`strip_face`
 Character. Font face for panel strip. Default: "bold"

`strip_text_size`
 Numeric. Font size for panel strip. Default: 16

`strip_background`
 Character. Background color of strip. Default: "lightgrey"

`strip_text_margin`
 Numeric vector c(top, right, bottom, left). Default: c(10, 0, 10, 0)

`panel_spacing`
 Numeric. Spacing between panels. Default: 2

`panel_rows`
 Numeric or NULL. Number of rows in panel layout. Default: NULL

`panel_cols`
 Numeric or NULL. Number of columns in panel layout. Default: NULL

`theme`
 ggplot2 theme or NULL. Custom ggplot theme. Default: NULL

`color_tone`
 Character or NULL. Base color theme. Default: NULL

`color_palette_type`
 Character. Type of color palette ('qualitative', 'sequential', 'diverging'). Default: "qualitative"

`positive_color`
 Character. Color for positive values. Default: "#2E8B57"

`negative_color`
 Character. Color for negative values. Default: "#CD5C5C"

`background_color`
 Character. Background color of plot. Default: "white"

grid_color	Character. Color of grid lines. Default: "grey90"
show_grid_major_x	Logical. Show major grid lines on x-axis. Default: FALSE
show_grid_major_y	Logical. Show major grid lines on y-axis. Default: FALSE
show_grid_minor_x	Logical. Show minor grid lines on x-axis. Default: FALSE
show_grid_minor_y	Logical. Show minor grid lines on y-axis. Default: FALSE
show_zero_line	Logical. Show or hide zero line. Default: TRUE
zero_line_type	Character. Line type ("solid", "dashed", "dotted"). Default: "dashed"
zero_line_color	Character. Color of zero line. Default: "black"
zero_line_size	Numeric. Line thickness of zero line. Default: 0.5
zero_line_position	Numeric. Position of the zero line. Default: 0
bar_width	Numeric. Width of bars. Default: 0.9
bar_spacing	Numeric. Spacing between groups of bars. Default: 0.9
scale_limit	Numeric vector of length 2 or NULL. Manual limits for value axis. Default: NULL
scale_increment	Numeric or NULL. Step size for axis tick marks. Default: NULL
expansion_y_mult	Numeric vector. Y-axis expansion. Default: c(0.05, 0.1)
expansion_x_mult	Numeric vector. X-axis expansion. Default: c(0.05, 0.05)
all_font_size	Numeric. Master control for all font sizes. Default: 1
sort_data_by_value	Logical. Whether to sort data by value. Default: FALSE
plot.margin	Numeric vector c(top, right, bottom, left). Margins around the entire plot. Default: c(10, 25, 10, 10)

Value

A list containing all style configuration parameters

Author(s)

Pattawee Puangchit

Examples

```
# Create customized style with title formatting
custom_style <- create_plot_style(
  color_tone = "gtap",
  title_size = 24,
  title_format = create_title_format(
    type = "prefix",
    text = "Impact on",
    sep = "-"
```

```

    ),
    bar_width = 0.5,
    x_axis_text_angle = 45
  )

```

create_title_format	Create a Title Format Configuration
---------------------	-------------------------------------

Description

Creates a configuration list for controlling plot title formatting. Supports auto-completion for common title format types.

Usage

```
create_title_format(type = "standard", text = "", sep = NULL)
```

Arguments

type	Character. Title format type: "standard": Default (variable + description + unit) "prefix": Adds text before the automatic title "suffix": Adds text after the automatic title "full": Uses only the specified text as the title "dynamic": Builds a title using column values
text	Character. Text content used for prefix, suffix, full, or a template for dynamic.
sep	Character. The separator between components (only used in "prefix" or "suffix" mode). Default is ":".

Value

A list with title format configuration parameters.

Author(s)

Pattawee Puangchit

Examples

```

# Standard auto-generated title
standard_title <- create_title_format()

# Prefix title
prefix_title <- create_title_format(
  type = "prefix",
  text = "Impact on",
  sep = " "
)

```



```
# Dynamic title using column values
dynamic_title <- create_title_format(
  type = "dynamic",
  text = "Impact on {Variable} in {Region}"
)
```

detail_plot	Create Comprehensive Bar Charts from HAR and SL4 Data
-------------	---

Description

Generates detailed bar charts to visualize the distribution of impacts across multiple dimensions. Supports top impact filtering, color coding, and fully customizable styling and export options.

Input Data

Usage

```
detail_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  split_by = "Variable",
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  invert_axis = TRUE,
  separate_figure = FALSE,
  top_impact = NULL,
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  plot_style_config = NULL
)
```

Arguments

data	A data frame or list of data frames containing GTAP results.
filter_var	NULL, a vector, a data frame, or a named list specifying filtering conditions. For example: list(Variable = c("EV", "qgdp"), REG = c("USA", "THA")).
x_axis_from	Character. Column name used for the x-axis.
split_by	Character or vector. • Column(s) used to split plots by (e.g., "REG" or c("COMM", "REG")). • If NULL, a single aggregated plot is generated.
panel_var	Character. Column for panel facets. Default is "Experiment".

variable_col Character. Column name for variable codes. Default is "Variable".
 unit_col Character. Column name for units. Default is "Unit".
 desc_col Character. Column name for variable descriptions. Default is "Description".

Plot Behavior

invert_axis Logical. If TRUE, flips the plot orientation (horizontal bars). Default is FALSE.
 separate_figure Logical. If TRUE, generates a separate plot for each value in panel_var. Default is FALSE.
 top_impact Numeric or NULL. If specified, shows only the top N impactful values; NULL shows all.

Variable Display

var_name_by_description Logical. If TRUE, uses descriptions instead of variable codes in titles. Default is FALSE.
 add_var_info Logical. If TRUE, appends variable codes in parentheses after the description. Default is FALSE.

Export Settings

output_path Character. Directory to save plots. If NULL, plots are returned but not saved.
 export_picture Logical. If TRUE, exports plots as PNG images. Default is TRUE.
 export_as_pdf Logical or "merged".
 • FALSE (default): disables PDF export.
 • TRUE: exports each plot as a separate PDF file.
 • "merged": combines all plots into a single PDF file.
 export_config List. Export options including dimensions, DPI, and background. See [create_export_config](#) or [get_all_config](#).

Styling

plot_style_config List. Custom plot appearance settings. See [create_plot_style](#) or [get_all_config](#).

Value

A ggplot object or a named list of ggplot objects depending on the separate_figure setting. If export_picture or export_as_pdf is enabled, the plots are also saved to output_path.

Author(s)

Pattawee Puangchit

See Also

[comparison_plot](#), [stack_plot](#)

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

# Prepare Dataframe
```

```

sector_data <- sl4.plot.data[["COMM*REG"]]

# Plot
plotB <- detail_plot(
  # === Input Data ===
  data      = sector_data,
  filter_var = list(Region = "Oceania"),
  x_axis_from = "Commodity",
  split_by   = "Region",
  panel_var   = "Experiment",
  variable_col = "Variable",
  unit_col    = "Unit",
  desc_col    = "Description",

  # === Plot Behavior ===
  invert_axis      = TRUE,
  separate_figure = FALSE,
  top_impact       = NULL,

  # === Variable Display ===
  var_name_by_description = TRUE,
  add_var_info            = FALSE,

  # === Export Settings ===
  output_path      = NULL,
  export_picture   = FALSE,
  export_as_pdf    = FALSE,
  export_config    = create_export_config(width = 45, height = 20),

  # === Styling ===
  plot_style_config = create_plot_style(
    positive_color = "#2E8B57",
    negative_color = "#CD5C5C",
    panel_rows = 1,
    panel_cols = NULL,
    show_axis_titles_on_all_facets = FALSE,
    y_axis_text_size = 25,
    bar_width = 0.6,
    all_font_size = 1.1
  )
)

```

get_all_config

*Print Plot and Export Configuration Snippets***Description**

Retrieve full configuration code as a list for applying in the plot styling and export settings.

Usage

```

get_all_config(
  plot_style = "default",
  plot_config = TRUE,
  export_config = TRUE
)

```

Arguments

plot_style Character. Plot style to use (currently only "default" is supported).
 plot_config Logical. If 'TRUE', prints the plot style configuration.
 export_config Logical. If 'TRUE', prints the export configuration.

Details

Once printing into the console, users can simply copy and paste the entire list of configurations, rename it (if needed), and use it in your plot functions directly.

Value

List of configuration snippets to the console.

Author(s)

Pattawee Puangchit

Examples

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

# Retrive configurations
get_all_config()

# Basic usage with data frame
p1 <- comparison_plot(
  data = sl4.plot.data[["REG"]],
  x_axis_from = "Region",
  panel_var = "Experiment",
  filter_var = c("qgdp", "EV"),
  output_path = NULL,
  export_picture = FALSE,
  export_as_pdf = FALSE,
  export_config = my_export_config,
  plot_style_config = my_style_config
)
```

get_color_palette *Print and Visualize Themed Color Palettes*

Description

Prints and visualizes predefined color palettes used in GTAPViz. Use 'color_tone = "all"' to return a list of callable palette functions.

Usage

```
get_color_palette(color_tone = NULL, palette_type = "qualitative")
```

Arguments

color_tone	Character. Name of the color theme to display (e.g., "gtap", "winter", "fall", or "all").
palette_type	Character. Palette type: "qualitative" (default), "sequential", or "diverging".

Author(s)

Pattawee Puangchit

Examples

```
# Get all palettes as callable functions
all_palettes <- get_color_palette("all")
all_palettes$winter()
all_palettes$gtap()

# Visualize specific palettes
get_color_palette("fall", "sequential")
get_color_palette("academic", "diverging")
```

`pivot_table_with_filter`

Export Data as an Excel Pivot Table

Description

Exports a dataset to an Excel file with both raw data and a generated pivot table.

Usage

```
pivot_table_with_filter(
  data,
  filter = NULL,
  rows = NULL,
  cols = NULL,
  data_fields = "Value",
  raw_sheet_name = "RawData",
  pivot_sheet_name = "PivotTable",
  dims = "A4",
  export_table = FALSE,
  output_path = NULL,
  workbook_name = "GTAP_PivotTable.xlsx"
)
```

Arguments

data	Data frame. The dataset to be exported.
filter	Character vector (optional). Columns to be used as filter fields in the pivot table.
rows	Character vector (optional). Columns to be used as row fields in the pivot table.

<code>cols</code>	Character vector (optional). Columns to be used as column fields in the pivot table.
<code>data_fields</code>	Character. The data field(s) to be summarized in the pivot table (default: "Value").
<code>raw_sheet_name</code>	Character. Name of the sheet containing raw data (default: "RawData").
<code>pivot_sheet_name</code>	Character. Name of the sheet containing the pivot table (default: "PivotTable").
<code>dims</code>	Character. Cell reference where the pivot table starts (default: "A3").
<code>export_table</code>	Logical. Whether to save the Excel file (default: TRUE).
<code>output_path</code>	Character. Directory where the file should be saved (default: current working directory).
<code>workbook_name</code>	Character. Name of the output Excel file (default: "GTAP_PivotTable.xlsx").

Details

This function creates an Excel workbook with:

- A raw data sheet (`raw_sheet_name`) containing the provided dataset.
- A pivot table sheet (`pivot_sheet_name`) generated based on specified row, column, and data fields.

If `export = TRUE`, the function saves the workbook to the specified `output_path`.

Value

An excel workbook object containing both raw data and the pivot table.

Author(s)

Pattawee Puangchit

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

data_pivot_table <- sl4.plot.data[["REG"]]

# Generate Pivot Table with Filter
pivot_table_with_filter(

  # === Input & Filter Settings ===
  data = data_pivot_table,
  filter = c("Variable", "Unit"), # Allow filtering by variable type and unit

  # === Pivot Structure ===
  rows = c("Region", "Sector"), # Rows: Regions and sectors
  cols = c("Experiment"),       # Columns: Experiments
  data_fields = "Value",        # Values to be aggregated

  # === Sheet & Layout ===
  raw_sheet_name = "Raw_Data",  # Sheet name for raw data
  pivot_sheet_name = "Sector_Pivot", # Sheet name for pivot table
```

```

    dims = "A3",                                # Starting cell for pivot table

    # === Export Options ===
    export = FALSE,
    output_path = NULL,
    workbook_name = "Sectoral_Impact_Analysis.xlsx"
  )
)
```

 rename_value

Rename Values in a Column

Description

Replaces specific values in a column based on a provided mapping file. Supports renaming across nested data structures and preserves factor levels.

Usage

```
rename_value(data, column_name = NULL, mapping.file)
```

Arguments

data	Data structure (data frame, list, or nested combination).
column_name	Character. Column to modify. If 'NULL', the function extracts it from 'mapping.file'.
mapping.file	Data frame with "OldName" and "NewName" columns for renaming.

Value

The same data structure with specified values replaced.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [convert_units](#), [sort_plot_data](#)

Examples

```

# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
har.plot.data <- readRDS(file.path(input_path, "har.plot.data.rds"))

# Rename variables in a dataset
mapping_welfare <- data.frame(
  ColumnName = "COLUMN",
  OldName = c("alloc_A1", "ENDWB1", "tech_C1", "pop_D1", "pref_G1", "tot_E1", "IS_F1"),
  NewName = c("Alloc Eff.", "Endwb", "Tech Chg.", "Pop", "Perf", "ToT", "I-S"),
  stringsAsFactors = FALSE
)
```

```
)

har.plot.data <- rename_value(har.plot.data, mapping.file = mapping_welfare)
```

report_table
Generate a Structured Report Table

Description

Transforms multiple datasets into wide-format tables based on defined pivot columns, hierarchical grouping, and renaming rules. Supports optional subtotal filtering and exporting to Excel.

Usage

```
report_table(
  data_list,
  pivot_col,
  total_column = FALSE,
  export_table = FALSE,
  separate_file = FALSE,
  output_path = NULL,
  sheet_names = NULL,
  include_units = FALSE,
  component_exclude = NULL,
  group_by = NULL,
  rename_cols = NULL,
  var_name_by_description = TRUE,
  add_var_info = FALSE,
  decimal = 2,
  unit_select = NULL,
  separate_sheet_by = NULL,
  subtotal_level = FALSE,
  repeat_label = FALSE,
  workbook_name = "detail_results",
  add_group_line = FALSE
)
```

Arguments

<code>data_list</code>	A named list of data frames to process.
<code>pivot_col</code>	A named list specifying the column to pivot into a wide format for each dataset. Each dataset can have only one pivot column. Example: <code>pivot_col = list(A = "COLUMN", E1 = "PRICES")</code>
<code>total_column</code>	Logical. If TRUE, adds a "Total" column summing numeric values.
<code>export_table</code>	Logical. If TRUE, saves the output as an Excel file.
<code>separate_file</code>	Logical. If TRUE, saves each dataset as a separate Excel file.
<code>output_path</code>	Character. Directory for saving Excel files when <code>export_table = TRUE</code> .
<code>sheet_names</code>	Optional named list for custom sheet names.

include_units	Logical. If TRUE, includes "Unit" as a grouping column if applicable.
component_exclude	Optional character vector specifying pivoted values to exclude.
group_by	A named list defining hierarchical grouping for each dataset. The order of columns in each list determines the priority. Example: <code>group_by = list(A = list("Experiment", "REG"), E1 = list("Experiment", "REG", "COMM"))</code>
rename_cols	A named list for renaming columns across all datasets. Example: <code>rename_cols = list("REG" = "Region", "COMM" = "Commodities", "Experiment" = "Scenario")</code>
var_name_by_description	Logical. If TRUE, replaces variable codes with descriptions when available.
add_var_info	Logical. If TRUE, appends variable codes in parentheses after descriptions.
decimal	Numeric. Number of decimal places for rounding values.
unit_select	Optional character. Specifies a unit to filter the dataset.
separate_sheet_by	Optional column name to split sheets in Excel. If defined, each unique value in the specified column gets its own sheet. Example: <code>separate_sheet_by = "Scenario"</code> .
subtotal_level	Logical. If TRUE, includes all subtotal values; otherwise, keeps only TOTAL rows.
repeat_label	Logical. If TRUE, repeats the first group column in exports for clarity.
workbook_name	Character. Name of the Excel workbook (without extension).
add_group_line	Logical. If TRUE, adds a thin line after each group in the exported table.

Details

This function requires a data list and can generate multiple output tables in a single setup. That is, all data frames within the list can be processed simultaneously. See the example for how to generate two data frames at once from the data list `sl4.plot.data`, which is obtained via `auto_gtap_ddta(plot_data = TRUE)`.

Value

If `export_table = TRUE`, tables are saved as Excel files.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [convert_units](#), [rename_value](#), [pivot_table_with_filter](#)

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

report_table(
  data_list = sl4.plot.data,

  # === Table Structure ===
```

```

pivot_col = list(
  REG = "Variable",
  COMM*REG = "Commodity"
),
group_by = list(
  REG = list("Experiment", "Region"),
  COMM*REG = list("Experiment", "Variable", "Region")
),
rename_cols = list("Experiment" = "Scenario"),

# === Table Layout & Labels ===
total_column = FALSE,
decimal = 4,
subtotal_level = FALSE,
repeat_label = FALSE,
include_units = TRUE,
var_name_by_description = TRUE,
add_var_info = TRUE,
add_group_line = FALSE,

# === Export Options ===
separate_sheet_by = "Unit",
export_table = FALSE,
output_path = NULL,
separate_file = FALSE,
workbook_name = "Comparison Table Default"
)

```

sort_plot_data

Sort GTAP Plot Data

Description

Sorts data frames in a GTAP plot list structure based on specified column orders. Works with data frames, lists of data frames, or nested data structures.

Usage

```

sort_plot_data(
  data,
  sort_columns = NULL,
  sort_by_value_desc = NULL,
  convert_to_factor = TRUE
)

```

Arguments

data A data frame or list structure containing data to be sorted.

sort_columns Named list. Specifies columns to sort by and their ordering. Each element should be a character vector of values in desired order. For example, `list(Region = c("USA", "EU", "Japan"), Experiment = c("Base", "Shock1", "Shock2"))`.

sort_by_value_desc

Logical or NULL. Controls sorting by the "Value" column: - NULL (default): Don't sort by value, only use column-based sorting. - TRUE: After column-based sorting, sort by value in descending order. - FALSE: After column-based sorting, sort by value in ascending order.

convert_to_factor

Logical. Whether to convert sorted columns to factors with custom ordering. Default is TRUE, which preserves ordering in GTAP plotting functions.

Value

A data structure with the same form as the input, with all contained data frames sorted.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [convert_units](#), [rename_value](#)

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

# Creating Sorting Rule
sorting_specs <- list(
  Experiment = c("EXP2", "EXP1"),      # Show EXP2 first, then EXP1
  Region = c("EastAsia", "SEAsia", "Oceania") # Custom region order
)

# Sorting
sort_data <- sort_plot_data(sl4_data1, sort_columns = sorting_specs,
                           sort_by_value_desc = FALSE)
```

stack_plot

Create Stacked Bar Charts for Decomposition Analysis

Description

Generates stacked bar charts to visualize value compositions across multiple dimensions. Supports both stacked and unstacked layouts for decomposition analysis, with full control over grouping, faceting, top-impact filtering, and export styling.

Input Data

Usage

```

stack_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  stack_value_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  invert_axis = FALSE,
  separate_figure = FALSE,
  show_total = TRUE,
  unstack_plot = FALSE,
  top_impact = NULL,
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  plot_style_config = NULL
)

```

Arguments

<code>data</code>	A data frame or list of data frames containing GTAP results.
<code>filter_var</code>	NULL, a vector, a data frame, or a named list specifying filtering conditions. For example: <code>list(Variable = c("EV", "qgdp"), REG = c("USA", "THA"))</code> .
<code>x_axis_from</code>	Character. Column name used for the x-axis.
<code>stack_value_from</code>	Character. Column containing stack component categories (e.g., "COMM" for commodities).
<code>split_by</code>	Character or vector. • Column(s) used to split plots by (e.g., "REG" or <code>c("COMM", "REG")</code>). • If NULL, a single aggregated plot is generated.
<code>panel_var</code>	Character. Column for panel facets. Default is "Experiment".
<code>variable_col</code>	Character. Column name for variable codes. Default is "Variable".
<code>unit_col</code>	Character. Column name for units. Default is "Unit".
<code>desc_col</code>	Character. Column name for variable descriptions. Default is "Description".
Plot Behavior	
<code>invert_axis</code>	Logical. If TRUE, flips the plot orientation (horizontal bars). Default is FALSE.
<code>separate_figure</code>	Logical. If TRUE, generates a separate plot for each value in <code>panel_var</code> . Default is FALSE.
<code>show_total</code>	Logical. If TRUE, displays total values above stacked bars. Default is TRUE.
<code>unstack_plot</code>	Logical. If TRUE, creates separate bar plots for each <code>x_axis_from</code> value instead of stacking. Default is FALSE.

top_impact Numeric or NULL. If specified, shows only the top N impactful values; NULL shows all.

Variable Display

var_name_by_description Logical. If TRUE, uses descriptions instead of variable codes in titles. Default is FALSE.

add_var_info Logical. If TRUE, appends variable codes in parentheses after the description. Default is FALSE.

Export Settings

output_path Character. Directory to save plots. If NULL, plots are returned but not saved.

export_picture Logical. If TRUE, exports plots as PNG images. Default is TRUE.

export_as_pdf Logical or "merged".

- FALSE (default): disables PDF export.
- TRUE: exports each plot as a separate PDF file.
- "merged": combines all plots into a single PDF file.

export_config List. Export options including dimensions, DPI, and background. See [create_export_config](#) or [get_all_config](#).

Styling

plot_style_config List. Custom plot appearance settings. See [create_plot_style](#) or [get_all_config](#).

Value

A ggplot object or a named list of ggplot objects depending on the `separate_figure` setting. If `export_picture` or `export_as_pdf` is enabled, the plots are also saved to `output_path`.

Author(s)

Pattawee Puangchit

See Also

[comparison_plot](#), [detail_plot](#)

Examples

```
# Load Data:
input_path <- system.file("extdata/in", package = "GTAPViz")
sl4.plot.data <- readRDS(file.path(input_path, "sl4.plot.data.rds"))

# Prepare Dataframe
welfare.decomp <- har.plot.data[["A"]]

# Plot
plotC <- stack_plot(
  # === Input Data ===
  data           = welfare.decomp,
  filter_var     = list(Region = "Oceania"),
  x_axis_from    = "Region",
  stack_value_from = "COLUMN",
  split_by      = FALSE,
  panel_var     = "Experiment",
```

```
variable_col      = "Variable",
unit_col          = "Unit",
desc_col          = "Description",

# === Plot Behavior ===
invert_axis       = FALSE,
separate_figure   = FALSE,
show_total        = TRUE,
unstack_plot      = FALSE,
top_impact        = NULL,

# === Variable Display ===
var_name_by_description = TRUE,
add_var_info       = FALSE,

# === Export Settings ===
output_path       = NULL,
export_picture     = FALSE,
export_as_pdf      = FALSE,
export_config      = create_export_config(width = 28, height = 15),

# === Styling ===
plot_style_config = create_plot_style(
  color_tone      = "gtap",
  panel_rows      = 2,
  panel_cols      = NULL,
  show_legend     = TRUE,
  show_axis_titles_on_all_facets = FALSE
)
)
```

Index

`add_mapping_info`, [2](#), [6](#), [10](#), [23](#), [25](#), [27](#)
`add_mapping_info()`, [4](#)
`auto_gtap_data`, [3](#)

`comparison_plot`, [6](#), [18](#), [29](#)
`convert_units`, [3](#), [5](#), [6](#), [9](#), [23](#), [25](#), [27](#)
`create_export_config`, [7](#), [10](#), [18](#), [29](#)
`create_plot_style`, [7](#), [11](#), [18](#), [29](#)
`create_title_format`, [8](#), [16](#)

`detail_plot`, [8](#), [17](#), [29](#)

`get_all_config`, [7](#), [8](#), [18](#), [19](#), [29](#)
`get_color_palette`, [20](#)
`group_data_by_dims`, [5](#)

`pivot_table_with_filter`, [21](#), [25](#)

`rename_value`, [3](#), [6](#), [10](#), [23](#), [25](#), [27](#)
`report_table`, [24](#)

`sort_plot_data`, [10](#), [23](#), [26](#)
`stack_plot`, [8](#), [18](#), [27](#)